

答案用紙

Examination Answer Sheet (提出年月日: submit Date

年 月 日)
Year Month Day

授業科目名 (担当教員名:) Course Name Instructor Name	学籍番号: 202318030 Student ID No	評点 Score
所 属: Bachelor's Program in School/College/Program Interdisciplinary Engineering	氏名: Andrea Trevanna Name	100

② (a) $\cdot x_n$; $N(AT)^T = N(A)$ because rows of A^T = columns of A

$$[A^T b] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \Rightarrow x_n = x_2 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\cdot x_p: \begin{cases} x_1 - x_2 - 2x_3 = 12 \\ x_2 \text{ free} \\ x_3 \text{ free} \end{cases} \rightarrow x_p = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} \text{ set free variables equal to zero}$$

$$\therefore x = x_n + x_p = x_2 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

$$(b) \cdot x_n: N(AT)^T = N(A) \rightarrow [A^T b] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \sim \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \sim \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$x_n = x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\cdot x_p: \begin{bmatrix} 1 & -1 & -2 & 12 \\ 0 & 1 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 8 \\ 0 & 1 & 1 & -4 \end{bmatrix} \rightarrow x_p = \begin{bmatrix} 8 \\ -4 \\ 0 \end{bmatrix}$$

$$\therefore x = x_n + x_p = x_3 \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 8 \\ -4 \\ 0 \end{bmatrix}$$

② No. of independent vectors of a (sub)space = dimension of a (sub)space

$$a) \dim C(A) = r = \text{no. of pivots} = 2 \quad b) \dim C(U) = r = 2$$

$$c) \dim C(A^T) = r = 2 \quad d) \dim C(U^T) = r = 2$$

The row space of A and the row space of U are the same $(c) = (d)$

$$③ (a) x = (A^T A)^{-1} A^T b = \left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$= \left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix}$$

$$\therefore p = Ax = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -\frac{2}{3} \\ -\frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$(b) x = (A^T A)^{-1} A^T b = \left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{2}{3} & \frac{2}{3} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -\frac{4}{3} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} \\ \frac{1}{6} \end{bmatrix}$$

$$\therefore p = Ax = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -\frac{2}{3} \\ \frac{1}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{6} \end{bmatrix}$$

$$④ a) \begin{bmatrix} C & D \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$$

To find least squares solution x : $A^T A x = A^T b \rightarrow x = (A^T A)^{-1} A^T b$

$$x = \left(\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \right)^{-1} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\therefore x = (C, D) = (2, 1)$$

$$b) \cdot v_1 = a = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\cdot v_2 = b - \hat{p}_{v_1}(b) = b - \frac{v_1^T b}{v_1^T v_1} v_1 = b - v_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow q_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$\cdot v_3 = c - \hat{p}_{v_2}(c) = c - \frac{v_2^T c}{v_2^T v_2} v_2 = c - v_2 - \frac{6}{8} v_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} - \frac{3}{4} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix} \rightarrow q_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix}$$

$$\therefore q_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, q_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \end{bmatrix}$$

□ 【裏面に続く】 (裏面を使用する場合は、チェックをすること)
Continued Overleaf If you continue overleaf please check the box

筑波大学
University of Tsukuba

(Answer for $A = QR$ is on next page)

答案用紙

Examination Answer Sheet (提出年月日 : submit Date

年 月 日)
Year Month Day

授業科目名 (担当教員名 : Course Name Instructor Name	学籍番号 : 202318030 Student ID No	評点 Score
所 属 : Bachelor's Program in School/College/Program Interdisciplinary Engineering	氏名 : Andrea Trevanno Name	

④ b) $A = QR = [q_1 \ q_2 \ q_3] \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix}$

• $q_1^T a = 1$ • $q_1^T b = 1$ • $q_1^T c = 1$ • $q_2^T b = 2\sqrt{2}$ • $q_2^T c = 3/\sqrt{2}$ • $q_3^T c = 1/\sqrt{2}$

$A = QR = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2\sqrt{2} & 3/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}$

Check = $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 2 \end{bmatrix} \checkmark$

Check No. 1

$$A = [a_1 \ a_2 \ a_3]$$

Name: Andrea Trevanna
Student ID: 202318030

$$1) \times 2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\text{Check if } 1a_1 + 1a_2 = 0 \quad \checkmark$$
$$1 - 1 = 0$$

$$\times 3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Check if } 2a_1 + a_3 = 0 \quad \checkmark$$
$$2 - 2 = 0$$

$$[1 \ -1 \ -2] \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} = 12 \quad \checkmark$$

$$b) \times 3 \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$A = [a_1 \ a_2 \ a_3]$$

$$\text{Check if } a_1 - a_2 + a_3 = \vec{0}$$

$$\rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \checkmark$$

For 1 (a): Check:

$$[1 \ -1 \ -2] \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix} = 12 \quad \checkmark$$

(b) Check:

$$\begin{bmatrix} 1 & -\frac{1}{2} & -2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 + (-4)(-1) + 0 \\ 1(-4) + 0 \end{bmatrix}$$
$$= \begin{bmatrix} 12 \\ -4 \end{bmatrix} \quad \checkmark$$

Check No. 4

$$q_1^T q_2 = [1 \ 0 \ 0] \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 0$$

$$q_1^T q_3 = [1 \ 0 \ 0] \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 0$$

$$q_2^T q_3 = [0 \ 1/\sqrt{2} \ 1/\sqrt{2}] \begin{bmatrix} 0 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 0$$

↪

Name: Andrea Trevanna
Student ID: 202318030

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2\sqrt{2} \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$
$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$1/\sqrt{2} (2\sqrt{2}) = 2$$

$$\frac{3}{2} - \frac{1}{2} = \frac{2}{2} = 1$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Name: Andren
Trevanna
Student ID: 202318030

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} = \frac{1}{6} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 12 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

(4) b)

$$\begin{bmatrix} 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 6$$

≥ 0

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{2}} 4 = \sqrt{2}$$

$$- " \quad \quad \quad \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \frac{3}{\sqrt{2}} \quad \quad \quad -1+2=$$

$$q_3^T c = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{2}}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2\sqrt{2} \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 3/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2\sqrt{2} \\ 0 \end{bmatrix} = 2$$

$$\begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{3}{2} - \frac{1}{2} = 1$$

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = 1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2\sqrt{2} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2\sqrt{2} \\ 0 \end{bmatrix} = 1/\sqrt{2}$$

? + ^

③ Check $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is in CCA : $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \checkmark$

Check $\begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$ is in CCA : $\begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix} = -2 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 6 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
 $= \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$

Name: Andrew Trevanna
Student ID: 202318030